

Quadratic approximation of determinant of deformation gradient in MPM

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We know from this MPM course and basically any MPM paper that:

$$F_p^{n+1} = (I + \Delta t \nabla \mathbf{v}) F_p^n$$

if we define $J^{n+1} = \det(F_p^{n+1})$ then $J^{n+1} = J^n \det(I + \Delta t \nabla \mathbf{v})$.

Note: From Taylor expansion we know $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$. Same is true for matrices, we can write $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$.

So we can write $I + \Delta t \nabla \mathbf{v} \approx e^{\Delta t \nabla \mathbf{v}}$. **Jacobi formula** tells us $\det(e^A) = e^{\text{tr}(A)}$, which can be further approximated using Taylor approximation as $e^{\text{tr}(A)} \approx 1 + \text{tr}(A) + \frac{\text{tr}(A)^2}{2!} + \dots$ so we can write:

$$\begin{aligned} J^{n+1} &= J^n \det(I + \Delta t \nabla \mathbf{v}) \\ &\approx J^n \det(e^{\Delta t \nabla \mathbf{v}}) \\ &= J^n e^{\text{tr}(\Delta t \nabla \mathbf{v})} \\ &\approx J^n \left(1 + \text{tr}(\Delta t \nabla \mathbf{v}) + \frac{\text{tr}(\Delta t \nabla \mathbf{v})^2}{2!} \right) \end{aligned}$$

in normal circumstances, we would just use $J^{n+1} = J^n \det(I + \Delta t \nabla \mathbf{v})$ but this approximation can be useful when J becomes inverted.

To be added: I forgot the proof that shows that adding the second term for approximation can help with inverted J . I will add it later.